Before Scientific Emergence Becomes Visible: Independent Evidence for Structural Precursors in Large-Scale Scholarly Knowledge

Mohammad Amir Khusru Akhtar

Usha Martin University, Ranchi – 834001, Jharkhand, India

Abstract

Can the organization of scientific knowledge carry detectable traces of future emergence before a research area becomes visibly successful? We test this question with PBSK, a temporally locked, reproducible science-of-science framework motivated by the seven-stage arc in *The Impossible Idea: How New Realities Enter the World*: possibility, question, pattern, imagination, insight, discovery, and transformation. The analysis separates a 12-landmark exploratory cohort from an independent power-validation cohort comprising 120 future-emergent OpenAlex topics and 240 pre-cutoff matched controls. All predictors are restricted to information available by 2018, while outcomes are defined over 2019–2023. In the independent cohort, pre-specified cross-domain collision was higher five years before emergence (paired $d = 0.266$, permutation $p = .0022$, BH-FDR $q = .011$), while sequence-order agreement was higher at both five years ($d = 0.217$, $q = .034$) and three years ($d = 0.219$, $q = .036$) before the outcome. Exploratory signals for question pressure, leap nonlinearity, and discovery readiness showed larger effects ($d \approx 0.37$–0.53, $q = .0012$ across several offsets) and are treated as hypothesis-generating. A conventional activity/citation baseline achieved ROC-AUC = .807 (95% CI [.765, .846]), exceeding the stage-only model (AUC = .713) and book-wide scalar score (AUC = .521). Adding the book-wide score or core geometry did not improve baseline discrimination; legacy PBS produced a small, statistically uncertain increment (ΔAUC = .0146, 95% CI [$-$.0085, .0372]). Thus, the evidence supports a narrower conclusion than "breakthrough prediction": future-emergent fields exhibit reproducible structural organization years before emergence, although these structural signals do not yet outperform conventional bibliometric activity for classification. The study provides a falsifiable computational operationalization, independent validation, and a fully reproducible OpenAlex/PubMed software pipeline.

Keywords: science of science; scientific emergence; breakthrough research; novelty; interdisciplinarity; knowledge graphs; OpenAlex; bibliometrics; prospective validation; reproducibility

1 Introduction

Scientific progress is strongly retrospective. Landmark discoveries are usually identified after recognition, citation, diffusion, prizes, or field formation has already occurred. Yet the science-of-science literature increasingly treats scientific development as a dynamical system whose future may be partially constrained by observable properties of the present (Clauset et al., 2017; Fortunato et al., 2018; Hou et al., 2025). Work on atypical combinations, recombinant search, interdisciplinary bridging, disruption, and knowledge diffusion shows that consequential science often depends on how previously separated ideas, communities, or methods interact (Foster et al., 2015; Kang et al., 2025; Kaplan & Vakili, 2015; Park et al., 2023; Schilling & Green, 2011; Shi & Evans, 2023; Uzzi et al., 2013). Recent machine-learning studies go further by forecasting new research directions from evolving knowledge graphs, scientific language models, and concept networks (Gu & Krenn, 2025; Luo et al., 2025; Marwitz et al., 2026). These advances motivate a sharper question: *before a scientific area becomes visibly emergent, does its knowledge structure already differ from comparable areas that will not undergo the same transition?*

The question is related to, but distinct from, forecasting highly cited papers. Citation-impact prediction has a mature literature (Acuna et al., 2012; Liu et al., 2018; Penner et al., 2013; Sinatra et al., 2016; D. Wang et al., 2013). Scientific emergence instead concerns the organization of a region of knowledge before the outcome is established. This distinction matters because the same bibliometric mechanisms that predict later impact can dominate any new structural score. A credible structural theory therefore must survive three tests: temporal locking, independent validation, and comparison against conventional activity/citation predictors.

This study develops and evaluates PBSK, a computational framework inspired by the conceptual sequence in *The Impossible Idea: How New Realities Enter the World* (Akhtar, 2026). The book's central proposition is treated here as a source of falsifiable hypotheses, not as empirical evidence. Its arc—possibility → question → pattern → imagination → insight → discovery → transformation—is translated into observable literature-level proxies. The resulting framework combines lexical signals, semantic geometry, topic diversity, cross-domain bridges, activity changes, and sequence ordering. Importantly, constructs such as "question," "imagination," and "insight" are not claimed to measure private cognition; they are proxies for properties of the published knowledge environment.

The design improves on a purely retrospective landmark analysis in four ways. First, a historical 12-landmark cohort is retained only as exploratory continuity. Second, the primary confirmatory analysis uses an independent OpenAlex topic sample: future-emergent topics are defined by post-2018 growth, while all predictors and matching variables are restricted to information available by 2018. Third, positive topics are matched to controls using pre-cutoff domain, volume, and momentum. Fourth, predictive models are evaluated with group-wise out-of-fold validation and group-bootstrap confidence intervals. This design is deliberately capable of rejecting the stronger theory.

The results do exactly that in part. Independent confirmatory tests support early cross-domain collision and ordered pre-emergence structure, while several exploratory stage signals are stronger still. Yet the full book-wide scalar score does not beat conventional bibliometric activity, and its addition does not improve a strong baseline. The paper therefore makes a deliberately narrower contribution: it identifies independently validated structural precursors of scientific emergence while showing that structural explanation and superior classification are not the same claim.

2 Related literature and theoretical positioning

2.1 Science as an evolving knowledge system

Science mapping has long represented the literature as networks of documents, citations, concepts, topics, and communities (Boyack et al., 2005; Klavans & Boyack, 2009; Newman, 2001; Rafols et al., 2010; van Eck & Waltman, 2010; Waltman & van Eck, 2012). Community structure and modularity provide a formal language for describing boundaries and bridges in these systems (Blondel et al., 2008; Newman, 2006; Traag et al., 2019). Topic models and modern language representations allow these boundaries to be estimated from text rather than citation links alone (Blei et al., 2003; Devlin et al., 2019; Reimers & Gurevych, 2019). Contemporary bibliometric infrastructures make such analyses feasible at very large scale, although source coverage and metadata quality remain consequential methodological choices (Alperin et al., 2024; Priem et al., 2022; Torres-Salinas & Arroyo-Machado, 2026; Visser et al., 2021).

The growth of science itself changes the search problem. As knowledge accumulates, researchers face increasing specialization and coordination burdens (Bloom et al., 2020; Jones, 2009). Team science has become increasingly dominant (Wuchty et al., 2007), while large and small teams appear to play different roles in development and disruption (Wu et al., 2019). Canonical progress may slow in mature, crowded fields (Chu & Evans, 2021), and electronic access can simultaneously widen availability while narrowing what is actually cited (Evans, 2008). These observations make "where new fields come from" a structural question as much as an individual creativity question.

2.2 Novelty, recombination, interdisciplinarity, and disruption

A central result across innovation studies is that novelty is often combinatorial. Atypical combinations of conventional knowledge are associated with high impact (Uzzi et al., 2013); risky strategies can bridge knowledge clusters but also face higher failure probabilities (Foster et al., 2015). Recombinant search can produce breakthrough ideas when distant components are joined in productive ways (Fleming, 2001; Fleming & Sorenson, 2004; Kaplan & Vakili, 2015; Schilling & Green, 2011). At the same time, novelty is not universally rewarded and can suffer delayed recognition or bibliometric bias (J. Wang et al., 2017).

Interdisciplinarity is similarly non-monotonic. Diversity can generate opportunities, but excessive cognitive distance may impose coordination and reception costs (Leahey et al., 2017; Porter & Rafols, 2009; Stirling, 2007; Yegros-Yegros et al., 2015). Recent large-scale evidence emphasizes surprising combinations that arise when outsiders connect distant scientific contexts (Shi & Evans, 2023). Emergence-detection studies have combined topic modeling and network measures to identify interdisciplinary areas before consolidation (Kim et al., 2024). These literatures motivate the PBSK collision, bridge, complementarity, and sequence constructs.

Disruption offers a related but different perspective: important work can destabilize dependence on predecessors rather than merely accumulate citations (Funk & Owen-Smith, 2017; Park et al., 2023). The decline of disruption over time has intensified interest in mechanisms that create qualitatively new trajectories. Yet disruption is itself an ex-post property. PBSK instead asks whether structural precursors can be measured before the outcome window.

2.3 Forecasting science and AI-assisted discovery

Forecasting in science spans individual careers, citation trajectories, grant outcomes, and topic evolution (Acuna et al., 2012; Clauset et al., 2017; Penner et al., 2013; Sinatra et al., 2016; D. Wang et al., 2013). Funding and peer-review studies further show that institutional selection can correlate with later productivity and impact, while also leaving room for substantial uncertainty (Azoulay et al., 2011, 2019; Jacob & Lefgren, 2011; Li & Agha, 2015). Recent work directly forecasts emerging topics from evolving knowledge graphs (Gu & Krenn, 2025), predicts neuroscience results with language models (Luo et al., 2025), and identifies candidate future materials-science combinations with large language models and concept graphs (Marwitz et al., 2026). AI is increasingly integrated into scientific discovery more broadly (Gao & Wang, 2024; H. Wang et al., 2023), and recent field-level work argues that new methods and instruments can trigger new scientific domains (Krauss, 2025).

These developments make it especially important to distinguish predictive accuracy from explanatory structure. A model may classify future emergence well because it recognizes growth, citation intensity, or volume. A structural theory makes a different claim: the organization of knowledge contains additional interpretable precursors. Our evaluation therefore reports both structural hypothesis tests and head-to-head predictive benchmarks.

3 Theory and computational operationalization

The conceptual source proposes that new realities move through an arc from latent possibility to transformation (Akhtar, 2026). We operationalize the six pre-discovery stages using observable scholarly proxies, summarized in table 1. The complete 40-chapter mapping generated by the software is preserved in Supplementary Table S2 and its machine-readable CSV.

Table 1: Stage-balanced operationalization of the book-derived pre-discovery arc.

Stage	Observable proxies	Aggregation
Possibility	possibility pressure; density gap; boundary activity	Geometric mean
Question	question pressure; anomaly pressure; doubt tension	Geometric mean
Pattern	pattern formation; analogical transfer; bridge formation; collision; complementarity	Geometric mean
Imagination	surprise geometry; originality risk; cross-domain similarity	Geometric mean
Insight	insight coherence; convergence; recombination	Geometric mean
Discovery readiness	leap nonlinearity; momentum; sequence order	Geometric mean

Note. These are literature-level proxies, not direct measurements of private cognitive states.

3.1 Semantic geometry and structured knowledge gaps

For region r at cutoff t, documents are represented by TF–IDF features followed by truncated SVD. A two-cluster representation yields semantic centroids $\boldsymbol{\mu}_1$ and $\boldsymbol{\mu}_2$. Let s_r be their cosine similarity. The density-gap proxy is

$$D_{r,t} = 1 - \frac{s_r + 1}{2}, \tag{1}$$

so larger values indicate greater separation between the two dominant semantic regions. Boundary activity is the fraction of documents whose difference in cosine similarity to the two centroids falls below a fixed margin $\tau = 0.12$:

$$B_{r,t} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}\left(|\cos(\mathbf{z}_i, \boldsymbol{\mu}_1) - \cos(\mathbf{z}_i, \boldsymbol{\mu}_2)| < \tau\right). \tag{2}$$

Topic complementarity $C_{r,t}$ is normalized Shannon entropy over topic assignments (Shannon, 1948). The Structured Knowledge Gap (SKG) is the geometric mean

$$\mathrm{SKG}_{r,t} = (D_{r,t} B_{r,t} C_{r,t})^{1/3}. \tag{3}$$

3.2 Collision, analogy, possibility, and temporal reorganization

The cross-domain collision index requires diversity, document-level topic bridging, and semantic separation:

$$\text{Collision}_{r,t} = C_{r,t}\, H_{r,t}\, \left(0.5 + 0.5 D_{r,t}\right), \tag{4}$$

where $H_{r,t}$ is the share of documents assigned to at least two distinct topics. Analogical transfer combines bridging, cross-domain semantic similarity $S_{r,t}$, and reference-based recombination $R_{r,t}$:

$$\text{Analogy}_{r,t} = H_{r,t} S_{r,t} \max(R_{r,t}, 0.05). \tag{5}$$

Possibility pressure combines gap geometry, complementarity, possibility-oriented language, and boundary activity:

$$P_{r,t} = 0.35 D_{r,t} + 0.25 C_{r,t} + 0.20 L_{r,t}^{(P)} + 0.20 B_{r,t}. \tag{6}$$

Question, anomaly, and doubt signals are bounded lexical rates, for example $Q_{r,t} = \tanh(4L_{r,t}^{(Q)})$ and $A_{r,t} = \tanh(5L_{r,t}^{(A)})$. These lexical variables are deliberately interpreted as properties of publication language rather than direct measures of scientists' mental states.

Semantic convergence compares two-cluster similarity in early versus recent windows. Surprise geometry is the mean cosine distance of recent documents from the earlier centroid, while insight coherence measures the increase in within-recent cohesion. Leap nonlinearity is the largest positive year-to-year change in log publication count, transformed to $[0, 1]$:

$$L_{r,t} = \frac{\tanh\left(\max_y \Delta \log(1 + n_{r,y})\right) + 1}{2}. \tag{7}$$

3.3 Sequence ordering and stage-balanced score

The expected observable proxy ordering is possibility $\rightarrow$ question $\rightarrow$ anomaly/doubt $\rightarrow$ collision/pattern $\rightarrow$ insight $\rightarrow$ leap. For each signal, an onset year is estimated from the first year exceeding its 70th percentile. The sequence-order score is the proportion of pairwise onsets concordant with the expected ordering, assigning 0.5 to ties:

$$O_{r,t} = \frac{1}{\binom{6}{2}} \sum_{i<j} \left[\mathbb{I}(\tau_i < \tau_j) + \frac{1}{2}\mathbb{I}(\tau_i = \tau_j) \right]. \tag{8}$$

Raw proxies are grouped into six stages. If $S_1, \ldots, S_6$ denote Possibility, Question, Pattern, Imagination, Insight, and Discovery Readiness, the book-wide score uses an equal-stage geometric mean:

$$\text{PBSK}_{r,t} = \left(\prod_{k=1}^{6} \max(S_k, \epsilon) \right)^{1/6}, \qquad \epsilon = 10^{-6}. \tag{9}$$

This prevents stages with more raw variables from dominating solely through feature count. A v3-motivated core geometry score combines SKG, collision, and sequence:

$$G_{r,t} = \left(\text{SKG}_{r,t}\, \text{Collision}_{r,t}\, O_{r,t}\right)^{1/3}. \tag{10}$$

Because this combination was motivated after inspection of earlier landmark results, it is exploratory in the landmark cohort and confirmatory only in the independent v4 cohort.

3.4 Transformation

Post-event transformation is represented by growth, topic diffusion, normalization language, and post-event citation adoption. The composite is

$$T_r = \frac{1}{4}\left(G_r^+ + D_r^+ + N_r^+ + A_r^+\right), \tag{11}$$

with post-event horizons censored at the current year (2026) rather than filled with nonexistent future observations.

4 Data and study design

4.1 Data sources

OpenAlex is the primary source because it provides openly accessible work metadata, citation counts, references, and topic assignments at scale (Priem et al., 2022). The implications of relying on a single bibliographic source are nontrivial: large bibliographic databases differ in coverage, citation-link completeness, discipline balance, and metadata quality (Alperin et al., 2024; Torres-Salinas & Arroyo-Machado, 2026; Visser et al., 2021). PubMed is therefore used as an independent biomedical source where queries and coverage allow comparison. The computational stack uses NumPy, SciPy, scikit-learn, and Matplotlib, whose foundational implementations are independently documented (Harris et al., 2020; Hunter, 2007; Pedregosa et al., 2011; Virtanen et al., 2020).

4.2 Two-cohort design

Table 2 summarizes the exploratory landmark cohort and the independent power-validation cohort, including their temporal roles and control structure.

Table 2: Study cohorts and temporal design.

Cohort	Positives	Controls	Temporal design	Inferential role
Exploratory landmark cohort	12	24	$T-5, T-3, T-1$	Historical landmark cases; v3-inspected signals are treated as exploratory.
Independent power-validation cohort	120	240	Cutoff 2018; outcomes 2019–2023	Future-emergent OpenAlex topics with controls matched using pre-cutoff information only.

Note. The independent cohort is the primary confirmatory analysis; the landmark cohort is retained for continuity and exploratory interpretation.

The landmark cohort contains 12 recognized historical cases with 24 controls and is analyzed at $T-5$, $T-3$, and $T-1$. Because earlier PBSK versions had already exposed these results, newly emphasized v4 signals are not treated as fresh confirmatory evidence in that cohort.

The primary validation cohort is constructed independently from OpenAlex topics. A sample of 800 topics with at least 1,000 works is screened. The information cutoff is 2018 and the outcome window is 2019–2023. For topic j, future growth is

$$Y_j = \log(1 + N_{j,2019:2023}) - \log(1 + N_{j,\leq 2018}). \tag{12}$$

Within-domain growth ranks reduce domination by rapidly expanding domains. The upper tail yields 120 positive topics. Each positive is matched to two controls using only pre-cutoff information. The matching distance is

$$d(i, j) = \left|\log(1 + V_i) - \log(1 + V_j)\right| + 0.5 \left|\log(1 + M_i) - \log(1 + M_j)\right|, \tag{13}$$

where V is pre-cutoff volume and M is pre-cutoff momentum. Future growth is used to define the outcome but never enters predictor construction or control matching.

4.3 Retrieval, sampling, and quality gate

The v4 software uses deterministic OpenAlex random sampling rather than the first N ranked results. Landmark windows allow up to 600 records; power-validation topic windows allow up to 300. The random seed is 20260816. Inferential rows require at least 60 documents, at least five publication years, and nonzero recent and prior activity. Sparse rows remain in QC outputs but are excluded from inferential modeling.

4.4 Statistical analysis

Positive-control contrasts are paired by matched group. One-sided Wilcoxon tests and 5,000-iteration paired permutation tests are reported, with Benjamini–Hochberg false-discovery-rate control within offset and hypothesis family (Benjamini & Hochberg, 1995). Effect size is paired Cohen's d. Predictive models use grouped out-of-fold cross-validation so that observations from one matched group cannot be split across training and test folds. Metrics include ROC-AUC (DeLong et al., 1988; Fawcett, 2006), PR-AUC (Saito & Rehmsmeier, 2015), Brier score, and balanced accuracy. Ninety-five percent confidence intervals and incremental AUC contrasts use 1,000 group-bootstrap iterations. The analysis includes a conventional activity/citation baseline, legacy PBS, book-wide PBSK, core geometry, six book stages, augmented baselines, full logistic theory, random forest, and histogram gradient boosting.

5 Results

5.1 Integrity, coverage, and independent validation

The frozen v4.0.0 run completed with status PASS, temporal-lock PASS, and power-validation integrity PASS. It generated 1,080 OpenAlex feature rows for 120 future-emergent topics and 240 matched controls, alongside 108 landmark OpenAlex rows and 48 landmark PubMed rows. The power-validation modeling contract contained both classes and all 120 matched groups. The software generated 34 result tables and 13 original diagnostic/analysis figures without figure-generation errors. Complete machine-readable outputs are included in the supplementary package.

5.2 Confirmatory structural evidence

Table 3 reports the complete pre-specified confirmatory contrast family for the independent validation cohort.

Table 3: Pre-specified confirmatory structural tests in the independent validation cohort.

Offset	Metric	Pairs	Mean diff.	d	Perm. p	BH-FDR q
5.000	skg	120.000	-0.031	-0.493	1.000	1.000
5.000	collision_index	120.000	0.014	0.266	0.002	0.011
5.000	sequence_order_score	120.000	0.034	0.217	0.014	0.034
5.000	bookwide_pbsk	120.000	-0.013	-0.124	0.912	1.000
5.000	core_geometry_score	120.000	0.005	0.080	0.194	0.323
3.000	skg	120.000	-0.024	-0.452	1.000	1.000
3.000	collision_index	120.000	0.004	0.075	0.210	0.380
3.000	sequence_order_score	120.000	0.032	0.219	0.007	0.036
3.000	bookwide_pbsk	120.000	-0.004	-0.048	0.700	0.875
3.000	core_geometry_score	120.000	0.004	0.067	0.228	0.380
1.000	skg	120.000	-0.029	-0.528	1.000	1.000
1.000	collision_index	120.000	3.00e-04	0.006	0.455	1.000
1.000	sequence_order_score	120.000	0.015	0.094	0.155	0.775
1.000	bookwide_pbsk	120.000	-0.003	-0.045	0.698	1.000
1.000	core_geometry_score	120.000	-0.006	-0.086	0.826	1.000

The strongest pre-specified confirmatory result is early cross-domain collision. At T–5, future-emergent topics had a higher collision index than matched controls (mean difference = .0138, paired d = .266, permutation p = .0022, BH-FDR q = .011). Sequence-order agreement was also higher at T–5 (difference = .0336, d = .217, q = .034) and $T-3$ (difference = .0315, d = .219, q = .036). These effects are small to modest but survive the pre-specified within-family multiplicity correction in an independent 120-group cohort. Figure 1 shows all pre-specified structural contrasts, including null and reversed effects.

Notably, SKG does not replicate the direction suggested by the earlier landmark exploration; its independent effect is not a positive confirmatory result. The book-wide scalar and core geometry are likewise not independently supported as universal positive contrasts. This divergence is scientifically important because it separates robust components from those that were specific to the smaller discovery cohort.

5.3 Exploratory stage signals

Table 4 summarizes the exploratory book-derived signals that survive within-family FDR correction.

Several exploratory constructs show larger effects than the pre-specified confirmatory variables. Question pressure is higher in future-emergent topics across all three offsets (d = .433 at $T-5$, .463 at $T-3$, and .494 at $T-1$; q = .0012 at each offset). Leap nonlinearity is also consistently higher (d = .502, .486, and .452), as is stage discovery readiness (d = .526, .528, and .372). Pattern formation becomes

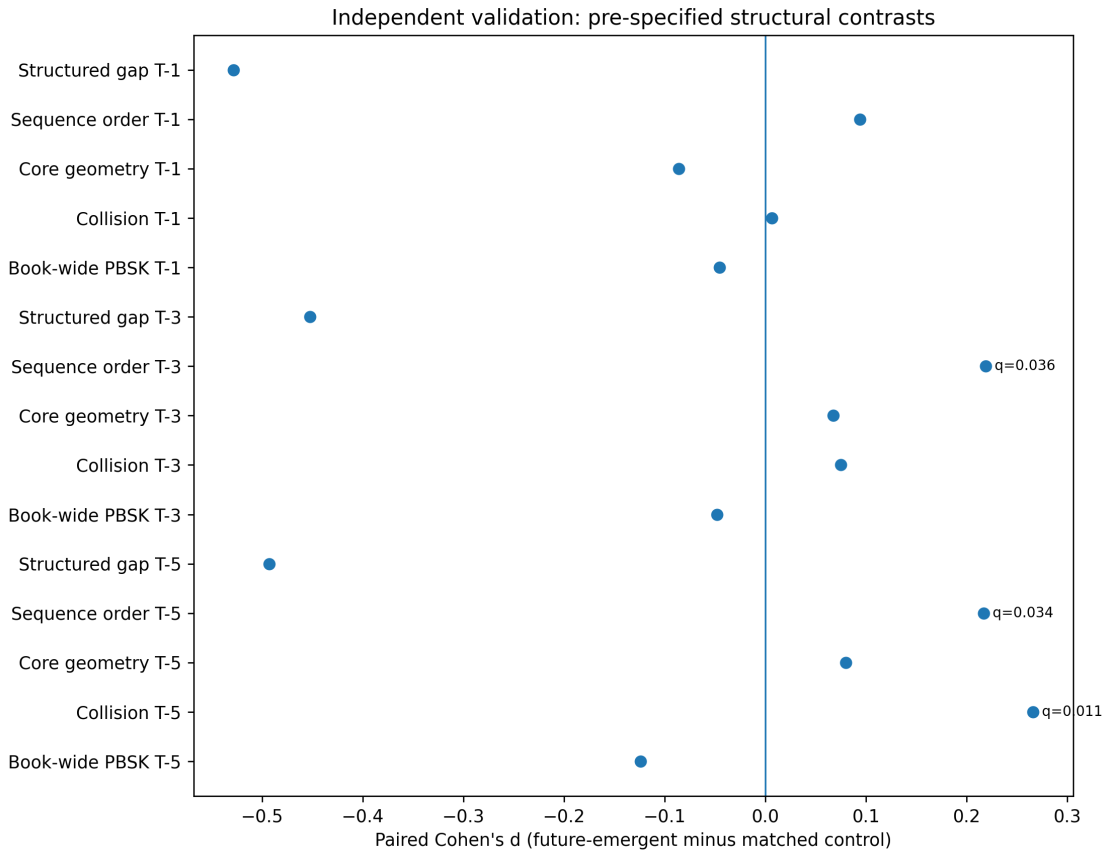

Figure 1: Independent validation of pre-specified structural contrasts. Points show paired Cohen's d for future-emergent topics minus matched controls. FDR-significant contrasts are annotated. The figure intentionally displays null and negative effects together with supported effects.

Table 4: Exploratory book-derived signals surviving within-family FDR correction.

Offset	Metric	Pairs	Mean diff.	d	Perm. p	BH-FDR q
5.000	question_pressure	120.000	0.075	0.433	2.00e-04	0.001
5.000	leap_nonlinearity	120.000	0.047	0.502	2.00e-04	0.001
5.000	stage_discovery_readiness	120.000	0.047	0.526	2.00e-04	0.001
3.000	question_pressure	120.000	0.077	0.463	2.00e-04	0.001
3.000	leap_nonlinearity	120.000	0.043	0.486	2.00e-04	0.001
3.000	stage_discovery_readiness	120.000	0.043	0.528	2.00e-04	0.001
1.000	question_pressure	120.000	0.078	0.494	2.00e-04	0.001
1.000	pattern_formation	120.000	0.004	0.266	0.002	0.009
1.000	leap_nonlinearity	120.000	0.038	0.452	2.00e-04	0.001
1.000	stage_discovery_readiness	120.000	0.034	0.372	2.00e-04	0.001

Note. These signals were not the pre-specified confirmatory targets and should be treated as hypothesis-generating.

significant at $T-1$ (d = .266, q = .009). These results suggest a potentially important Question–Leap–Readiness pattern, but because these variables were not the pre-specified confirmatory targets of the v4 power validation, they should seed a new untouched validation rather than be retroactively promoted to confirmation.

5.4 Prediction: strong baseline, limited incremental value

Table 5 reports grouped out-of-sample predictive performance for all candidate models in the independent validation cohort.

Table 5: Grouped out-of-sample predictive performance in the independent validation cohort.

Model	ROC-AUC	CI low	CI high	PR-AUC	Brier	Bal. acc.
Activity/citation baseline	0.807	0.765	0.846	0.717	0.178	0.749
Legacy PBS only	0.435	0.385	0.489	0.312	0.251	0.456
Book-wide PBSK only	0.521	0.464	0.571	0.352	0.251	0.542
Core geometry only	0.458	0.402	0.515	0.313	0.251	0.458
Book stages	0.713	0.667	0.757	0.585	0.213	0.655
Baseline + legacy PBS	0.822	0.781	0.856	0.729	0.170	0.748
Baseline + book-wide PBSK	0.805	0.764	0.844	0.708	0.178	0.744
Baseline + core geometry	0.806	0.764	0.844	0.713	0.178	0.739
Full theory logistic	0.798	0.754	0.837	0.717	0.178	0.733
Random forest full	0.791	0.751	0.831	0.710	0.171	0.711
HistGradientBoosting full	0.799	0.757	0.838	0.715	0.181	0.717

Predictive benchmarking sharply constrains the interpretation. The activity/citation baseline achieves ROC-AUC = .807 (95% CI [.765, .846]) and PR-AUC = .717. The six book stages achieve AUC = .713, showing that stage information contains nontrivial discriminative structure but substantially less than the conventional baseline. The scalar book-wide PBSK score is near chance (AUC = .521), while core geometry is below chance in this orientation (AUC = .458). A baseline augmented with legacy PBS reaches AUC = .822, but the incremental contrast is small and statistically uncertain.

Figure 2 visualizes the same grouped out-of-sample discrimination and makes the relative ranking of the conventional baseline and structural models explicit.

Table 6 quantifies the incremental predictive value of each pre-specified augmentation relative to the activity/citation baseline.

The incremental-value tests make the negative result explicit. Adding book-wide PBSK to the activity/citation baseline changes AUC by −0.0020 (95% CI [−.0046, .0007]); adding core geometry changes AUC by −0.0013 (CI [−.0087, .0065]). Legacy PBS provides ΔAUC = .0146, but its interval crosses zero ([−.0085, .0372]; one-sided bootstrap p = .106). Thus, the current evidence does not support the claim that PBSK improves classification beyond a strong conventional baseline.

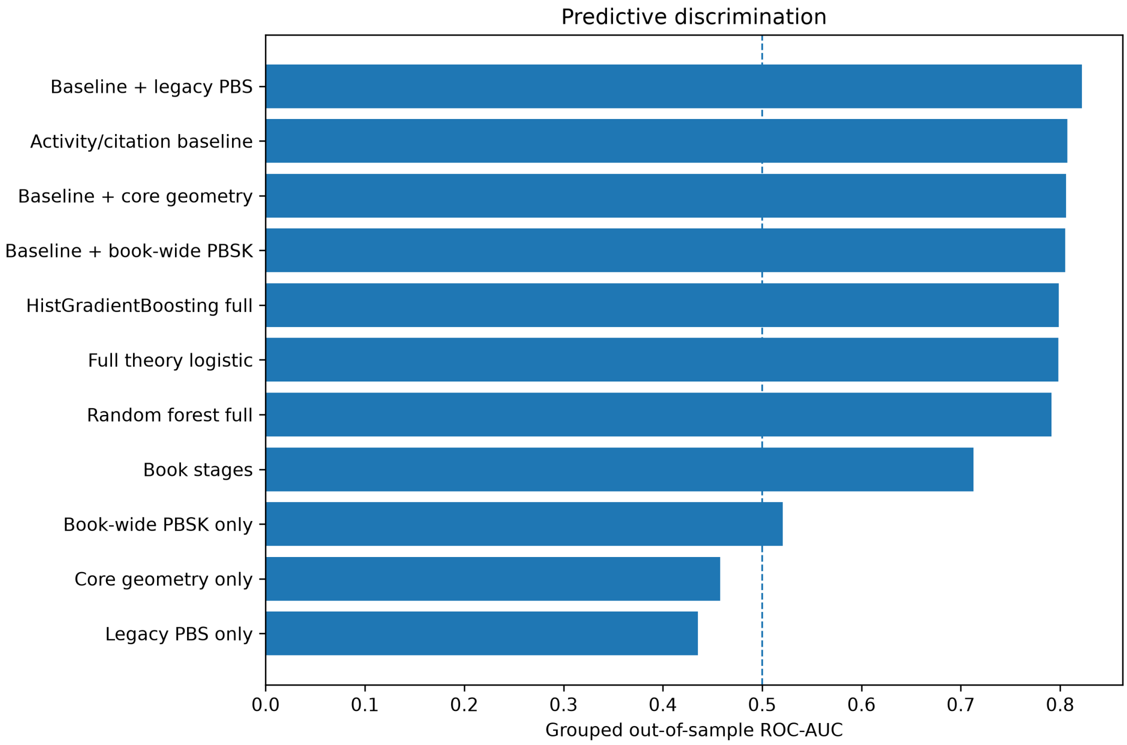

Figure 2: Grouped out-of-sample predictive discrimination in the independent power-validation cohort, reproduced directly from the frozen PBSK v4.0.0 output. The dashed line marks chance ROC-AUC = .5.

Table 6: Incremental predictive value relative to the activity/citation baseline.

Augmented model	Delta AUC	CI low	CI high	Bootstrap p
Baseline + legacy PBS	0.015	-0.008	0.037	0.106
Baseline + book-wide PBSK	-0.002	-0.005	6.84e-04	0.932
Baseline + core geometry	-0.001	-0.009	0.007	0.635
Full theory logistic	-0.009	-0.040	0.024	0.675

Note. Positive delta AUC favors the augmented model. None of the pre-specified augmentations provides a statistically convincing improvement over the conventional baseline.

5.5 Temporal discrimination

Table 7 reports predictive discrimination separately at $T-5$, $T-3$, and $T-1$.

Table 7: Temporal predictive performance at T-5, T-3, and T-1.

Offset	Model	ROC-AUC	CI low	CI high	PR-AUC	Brier
5.000	Activity/citation baseline	0.816	0.768	0.850	0.722	0.172
5.000	Book-wide PBSK only	0.538	0.479	0.578	0.370	0.251
5.000	Core geometry only	0.509	0.458	0.560	0.355	0.250
5.000	Baseline + core geometry	0.811	0.762	0.845	0.715	0.174
5.000	Full theory logistic	0.799	0.763	0.836	0.717	0.179
3.000	Activity/citation baseline	0.820	0.771	0.855	0.739	0.169
3.000	Book-wide PBSK only	0.516	0.464	0.562	0.343	0.251
3.000	Core geometry only	0.507	0.455	0.559	0.327	0.251
3.000	Baseline + core geometry	0.818	0.767	0.854	0.736	0.170
3.000	Full theory logistic	0.819	0.785	0.854	0.760	0.168
1.000	Activity/citation baseline	0.828	0.786	0.864	0.750	0.166
1.000	Book-wide PBSK only	0.509	0.459	0.553	0.344	0.251
1.000	Core geometry only	0.526	0.468	0.599	0.360	0.250
1.000	Baseline + core geometry	0.829	0.785	0.867	0.748	0.165
1.000	Full theory logistic	0.818	0.780	0.857	0.736	0.169

The conventional baseline strengthens modestly from AUC = .816 at $T-5$ to .828 at $T-1$. Full theory logistic reaches .799, .819, and .818 across the same offsets, while book-wide PBSK remains close to chance. Figure 3 emphasizes that the structural framework's strongest contribution is not superior temporal classification.

5.6 Ablation and stage dependence

Table 8 reports the change in predictive performance after removing each stage or composite component.

Table 8: Stage ablation analysis for the independent validation cohort.

Removed component	ROC-AUC	PR-AUC	Brier	Delta AUC
none	0.803	0.711	0.177	0.000
stage_possibility	0.790	0.671	0.184	-0.013
stage_question	0.807	0.717	0.175	0.005
stage_pattern	0.805	0.714	0.176	0.002
stage_imagination	0.802	0.706	0.177	-3.94e-04
stage_insight	0.804	0.713	0.177	0.001
stage_discovery_readiness	0.804	0.714	0.176	9.07e-04
bookwide_pbsk	0.806	0.713	0.176	0.004
core_geometry_score	0.804	0.714	0.176	0.001

Note. Negative delta indicates reduced discrimination relative to the full stage-based model.

Removing the Possibility stage reduces ROC-AUC by approximately .013, the largest deterioration

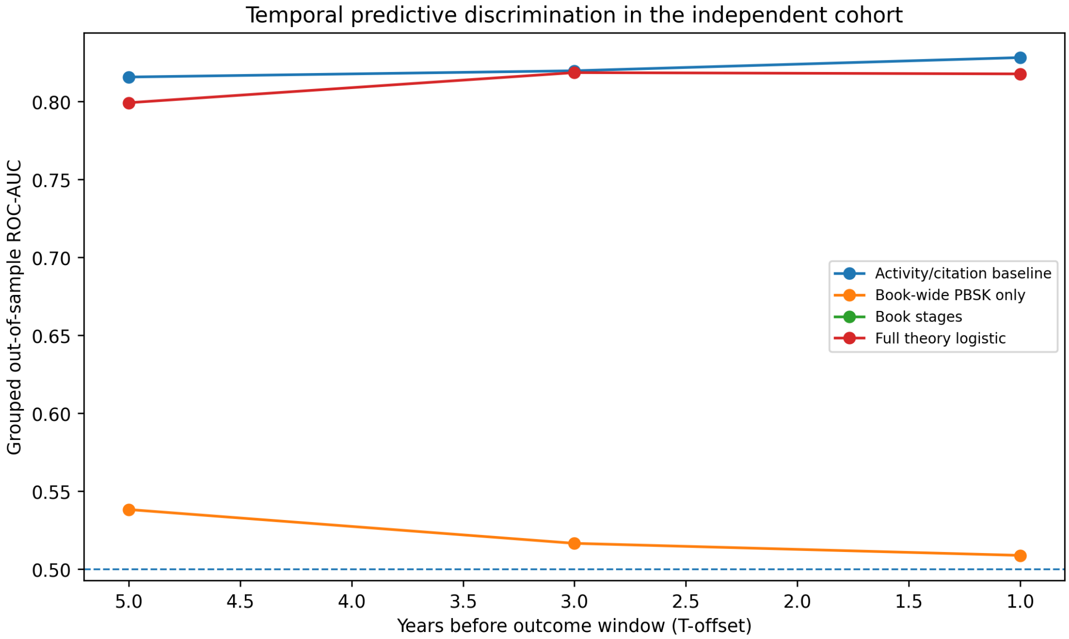

Figure 3: Temporal predictive discrimination in the independent cohort. Values are generated from the software's grouped out-of-fold results at $T-5$, $T-3$, and $T-1$.

among the stage ablations. Removing the other individual stages yields minimal change or slight improvement. This pattern is consistent with redundancy among the stage proxies and argues against interpreting the six-stage composite as six independently necessary causal ingredients. The ablation result is therefore informative precisely because it prevents an overly literal reading of the conceptual arc.

5.7 Historical landmark continuity and transformation

The smaller landmark cohort remains useful as an exploratory stress test but has wide uncertainty. The random-forest full model reaches AUC = .657 (95% CI [.503, .813]), whereas the activity/citation baseline is .559. None of the landmark incremental contrasts provides stable evidence of added value. The sequence architecture in the landmark cohort is mixed across offsets, which further justifies privileging the independently sampled power-validation results.

Table 9 reports post-landmark growth, diffusion, normalization, adoption, and the composite transformation contrast.

Table 9: Post-landmark transformation tests in the historical cohort.

Metric	Pairs	Mean diff.	d	Perm. p	BH-FDR q
post_growth	12.000	0.064	0.455	0.074	0.184
topic_diffusion	12.000	-0.039	-0.799	0.994	0.994
normalization_index	12.000	0.058	0.508	0.055	0.184
post_adoption	12.000	-0.004	-0.037	0.546	0.682
transformation	12.000	0.020	0.298	0.163	0.271

Note. No transformation metric survives FDR correction.

Post-landmark transformation also remains unresolved. Growth and normalization trend positively, but no transformation metric survives FDR correction. Therefore, the empirical support in this paper concerns *pre-emergence structure*, not a validated post-discovery transformation law.

5.8 Cross-source evidence and its limits

Table 10 summarizes descriptive agreement between OpenAlex and PubMed features in the available power-validation overlap.

Table 10: Cross-source agreement for OpenAlex and PubMed features in the power-validation overlap.

Metric	n	Pearson r	Spearman rho	Mean abs. diff.
bookwide_pbsk	79.000	0.100	-0.047	0.363
core_geometry_score	79.000	-0.027	-0.046	0.162
skg	79.000	0.043	-0.020	0.165
possibility_pressure	79.000	0.078	0.140	0.125
question_pressure	79.000	0.111	0.347	0.253
collision_index	79.000	-0.009	-0.024	0.154
insight_coherence	79.000	0.046	0.004	0.416
sequence_order_score	79.000	0.168	0.148	0.137

Note. Agreement is descriptive because the v4 PubMed confirmatory hypothesis table contains no valid replicated test family.

PubMed supplies a useful source-sensitivity check but not a complete independent confirmatory replication of the 120-topic OpenAlex test. The frozen v4 run reports zero valid PubMed power-validation hypothesis tests, while cross-source agreement is available for 79 overlapping region observations. Agreement is weak for most metrics (for example, Spearman $\rho = .347$ for question pressure and .148 for sequence order). This source dependence is a substantive limitation and argues for future replication using independently harmonized biomedical topic definitions rather than treating OpenAlex and PubMed features as interchangeable.

6 Discussion

6.1 What is supported

The strongest conclusion is structural, not predictive. Two pre-specified signals survive independent FDR-controlled testing: cross-domain collision at $T-5$, and ordered sequence structure at $T-5$ and $T-3$. This result aligns with literatures that connect high-impact novelty to atypical combinations, distant disciplinary interaction, and recombination (Foster et al., 2015; Kaplan & Vakili, 2015; Schilling & Green, 2011; Shi & Evans, 2023; Uzzi et al., 2013). It also complements work on interdisciplinary emergence (Kim et al., 2024) and on field formation through new methodological capabilities (Krauss, 2025). The effect sizes are not large, but they are detected years before the outcome window in a cohort not used to generate the earlier landmark observations.

This matters because scientific emergence need not be a singular "eureka" event. The data are compatible with a weaker, testable proposition: some future-emergent regions become structurally unusual before their later growth is obvious. Early collision suggests that interaction among diverse knowledge areas may precede consolidation; sequence-order support suggests that the temporal arrangement of observable signals carries information beyond their instantaneous levels.

6.2 What is not supported

The paper does not establish a superior breakthrough-prediction algorithm. Conventional activity/citation features classify the outcome much better than the scalar book-wide PBSK score. Augmenting the baseline with book-wide PBSK or core geometry does not improve ROC-AUC. The stronger claim—that the full conceptual architecture provides predictive information beyond conventional bibliometrics—is therefore rejected for this frozen experiment.

This negative result is not an implementation failure. It is a substantive boundary condition. Scientific growth outcomes are naturally correlated with pre-existing activity and citations, so a baseline containing those variables is difficult to beat. A structural framework may still explain *how* knowledge regions reorganize even when it does not improve ranking accuracy. Conflating explanation with prediction would overstate the evidence.

6.3 Why the exploratory Question–Leap–Readiness pattern deserves a new test

Question pressure, leap nonlinearity, and discovery readiness show the largest and most consistent exploratory effects in the independent cohort. Their recurrence across $T-5$, $T-3$, and $T-1$ suggests a potentially more parsimonious mechanism than the full scalar score: future-emergent fields may accumulate explicit problem/question language and nonlinear activity shifts before the later growth outcome. Yet using the same cohort to discover and confirm this pattern would be circular. The appropriate next step is an untouched preregistered validation cohort with these variables declared in advance.

6.4 Implications for science policy and AI-assisted discovery

If replicated, structural precursor measures could complement rather than replace conventional indicators. Funding systems already confront a tension between rewarding established productivity and supporting risky novelty (Azoulay et al., 2011; Jacob & Lefgren, 2011; Li & Agha, 2015; J. Wang et al., 2017). A structural dashboard might identify regions where distant communities are beginning to interact even before citation-based momentum becomes dominant. This would be most useful as a hypothesis-generation instrument, not an automated funding rule.

The same distinction applies to AI-assisted science. Models can increasingly forecast outcomes and propose candidate research directions (Gu & Krenn, 2025; Luo et al., 2025; Marwitz et al., 2026; H. Wang et al., 2023), but high predictive accuracy does not by itself establish a mechanism of discovery. PBSK offers an interpretable test bed in which machine learning, bibliometrics, and theory-derived proxies can be evaluated against one another under temporal constraints.

7 Robustness, reproducibility, and scientific guardrails

The software was designed so a successful workflow means computational integrity, not theoretical confirmation. No p-value, AUC, effect size, or favorable conclusion is hard-coded. Queries, seeds, cutoffs, manifests, and generated tables are preserved. The temporal leakage audit passes for the frozen run. Grouped cross-validation prevents matched observations from leaking across folds. The quality gate excludes sparse regions from inference. Multiple-testing control separates confirmatory from exploratory families rather than allowing a large exploratory family to define the confirmatory threshold.

The computational environment is fully reproducible from the public repository https://github.com/Arithmetic-Power-Geometry/PBSK; the software is released under Apache License 2.0. All 34 generated result tables, all 13 original software figures, run summaries, modeling contracts, and a snapshot of the analysis code/configuration used to produce this manuscript are included in the submission package.

8 Limitations

First, the independent outcome is *future topic growth*, not a gold-standard label of scientific breakthrough. The paper therefore studies emergence and should not equate rapid field growth with epistemic importance. Second, OpenAlex topic definitions and metadata determine the unit of analysis. Open bibliographic coverage is broad but not error-free, and source-specific metadata can affect inferred structure (Alperin et al., 2024; Torres-Salinas & Arroyo-Machado, 2026; Visser et al., 2021). Third, lexical proxies cannot observe curiosity, doubt, imagination, or insight inside scientists' minds; they only characterize scholarly language. Fourth, matching on domain, volume, and momentum reduces but cannot eliminate confounding. Fifth, the stage score imposes equal stage weights and a geometric aggregation by theory rather than estimating a latent-variable measurement model. Sixth, PubMed cross-source replication is incomplete for the new confirmatory cohort, and observed metric agreement is weak. Seventh, the 2018/2019–2023 cutoff is one historical slice. Replication over multiple non-overlapping cutoffs is required to establish temporal generality. Finally, the exploratory Question–Leap–Readiness pattern needs a new untouched validation before it can be called replicated.

9 Conclusion

Scientific emergence is neither fully predictable nor wholly invisible in advance. In an independent cohort of 120 future-emergent topics and 240 matched controls, cross-domain collision and book-derived sequence ordering show small but FDR-significant differences years before the outcome window. Several additional stage proxies exhibit stronger exploratory effects. At the same time, conventional activity/citation variables remain substantially better classifiers, and the book-wide scalar score adds no convincing incremental predictive value. The evidence therefore supports a restrained but nontrivial claim: *future-emergent scientific regions can exhibit measurable structural precursors before emergence, even when those precursors are not superior forecasting features*. This separation of structural explanation from predictive performance is the central empirical contribution and provides a clear path for preregistered, multi-source replication.

Data and code availability

The complete PBSK software is available at https://github.com/Arithmetic-Power-Geometry/PBSK under Apache License 2.0. The submission package contains the frozen v4.0.0 result tables, figures, run summary, modeling contracts, and analysis configuration used in this manuscript. OpenAlex and PubMed remain the upstream scholarly data providers; the repository retrieves only the records needed for the configured experiment and caches query manifests for reproducibility.

Ethics statement

The study analyzes bibliographic metadata and publication text. It involves no recruitment, intervention, or private human-subject data.

Competing interests

The author declares no competing interests.

Author contributions

Mohammad Amir Khusru Akhtar: conceptualization, theory development, methodology, software design, computational analysis, validation design, writing, visualization, and project administration.

References

Acuna, D. E., Allesina, S., & Kording, K. P. (2012). Future impact: Predicting scientific success. *Nature*, *489*, 201–202. https://doi.org/10.1038/489201a

Akhtar, M. A. K. (2026). *The impossible idea: How new realities enter the world* [Published under the pen name Shunya]. https://a.co/d/03buY0qI

Alperin, J. P., Portenoy, J., Demes, K., Lariviere, V., & Haustein, S. (2024). An analysis of the suitability of openalex for bibliometric analyses. *arXiv*. https://doi.org/10.48550/arXiv.2404.17663

Azoulay, P., Fons-Rosen, C., & Graff Zivin, J. S. (2019). Does science advance one funeral at a time? *American Economic Review*, *109*(8), 2889–2920. https://doi.org/10.1257/aer.20161574

Azoulay, P., Graff Zivin, J. S., & Manso, G. (2011). Incentives and creativity: Evidence from the academic life sciences. *The RAND Journal of Economics*, *42*(3), 527–554. https://doi.org/10.1111/j.1756-2171.2011.00140.x

Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, *57*(1), 289–300. https://doi.org/10.1111/j.2517-6161.1995.tb02031.x

Blei, D. M., Ng, A. Y., & Jordan, M. I. (2003). Latent dirichlet allocation. *Journal of Machine Learning Research*, *3*, 993–1022.

Blondel, V. D., Guillaume, J.-L., Lambiotte, R., & Lefebvre, E. (2008). Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment*, *2008*(10), P10008. https://doi.org/10.1088/1742-5468/2008/10/P10008

Bloom, N., Jones, C. I., Van Reenen, J., & Webb, M. (2020). Are ideas getting harder to find? *American Economic Review*, *110*(4), 1104–1144. https://doi.org/10.1257/aer.20180338

Boyack, K. W., Klavans, R., & Borner, K. (2005). Mapping the backbone of science. *Scientometrics*, *64*(3), 351–374. https://doi.org/10.1007/s11192-005-0255-6

Chu, J. S. G., & Evans, J. A. (2021). Slowed canonical progress in large fields of science. *Proceedings of the National Academy of Sciences*, *118*(41), e2021636118. https://doi.org/10.1073/pnas.2021636118

Clauset, A., Larremore, D. B., & Sinatra, R. (2017). Data-driven predictions in the science of science. *Science*, *355*(6324), 477–480. https://doi.org/10.1126/science.aal4217

DeLong, E. R., DeLong, D. M., & Clarke-Pearson, D. L. (1988). Comparing the areas under two or more correlated receiver operating characteristic curves: A nonparametric approach. *Biometrics*, *44*(3), 837–845. https://doi.org/10.2307/2531595

Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). Bert: Pre-training of deep bidirectional transformers for language understanding. *Proceedings of NAACL-HLT 2019*, 4171–4186. https://doi.org/10.18653/v1/N19-1423

Evans, J. A. (2008). Electronic publication and the narrowing of science and scholarship. *Science*, *321*(5887), 395–399. https://doi.org/10.1126/science.1150473

Fawcett, T. (2006). An introduction to roc analysis. *Pattern Recognition Letters*, *27*(8), 861–874. https://doi.org/10.1016/j.patrec.2005.10.010

Fleming, L. (2001). Recombinant uncertainty in technological search. *Management Science*, *47*(1), 117–132. https://doi.org/10.1287/mnsc.47.1.117.10671

Fleming, L., & Sorenson, O. (2004). Science as a map in technological search. *Strategic Management Journal*, *25*(8-9), 909–928. https://doi.org/10.1002/smj.384

Fortunato, S., Bergstrom, C. T., Borner, K., Evans, J. A., Helbing, D., Milojevic, S., Petersen, A. M., Radicchi, F., Sinatra, R., Uzzi, B., Vespignani, A., Waltman, L., Wang, D., & Barabasi, A.-L. (2018). Science of science. *Science*, *359*(6379), eaao0185. https://doi.org/10.1126/science.aao0185

Foster, J. G., Rzhetsky, A., & Evans, J. A. (2015). Tradition and innovation in scientists' research strategies. *American Sociological Review*, *80*(5), 875–908. https://doi.org/10.1177/0003122415601618

Funk, R. J., & Owen-Smith, J. (2017). A dynamic network measure of technological change. *Management Science*, *63*(3), 791–817. https://doi.org/10.1287/mnsc.2015.2366

Gao, J., & Wang, D. (2024). Quantifying the use and potential benefits of artificial intelligence in scientific research. *Nature Human Behaviour*, *8*, 2281–2292. https://doi.org/10.1038/s41562-024-02020-5

Gu, X., & Krenn, M. (2025). Forecasting high-impact research topics via machine learning on evolving knowledge graphs. *Machine Learning: Science and Technology*, *6*(2), 025041. https://doi.org/10.1088/2632-2153/add6ef

Harris, C. R., Millman, K. J., van der Walt, S. J., et al. (2020). Array programming with numpy. *Nature*, *585*, 357–362. https://doi.org/10.1038/s41586-020-2649-2

Hou, J., Zheng, B., Li, H., & Li, W. (2025). Evolution and impact of the science of science: From theoretical analysis to digital-ai driven research. *Humanities and Social Sciences Communications*, *12*, 316. https://doi.org/10.1057/s41599-025-04617-1

Hunter, J. D. (2007). Matplotlib: A 2d graphics environment. *Computing in Science & Engineering*, *9*(3), 90–95. https://doi.org/10.1109/MCSE.2007.55

Jacob, B. A., & Lefgren, L. (2011). The impact of research grant funding on scientific productivity. *Journal of Public Economics*, *95*(9-10), 1168–1177. https://doi.org/10.1016/j.jpubeco.2011.05.005

Jones, B. F. (2009). The burden of knowledge and the death of the renaissance man: Is innovation getting harder? *The Review of Economic Studies*, *76*(1), 283–317. https://doi.org/10.1111/j.1467-937X.2008.00531.x

Kang, D., Danziger, R. S., Rehman, J., & Evans, J. A. (2025). Limited diffusion of scientific knowledge forecasts collapse. *Nature Human Behaviour*, *9*(2), 268–276. https://doi.org/10.1038/s41562-024-02041-0

Kaplan, S., & Vakili, K. (2015). The double-edged sword of recombination in breakthrough innovation. *Strategic Management Journal*, *36*(10), 1435–1457. https://doi.org/10.1002/smj.2294

Kim, K., Kogler, D. F., & Maliphol, S. (2024). Identifying interdisciplinary emergence in the science of science: Combination of network analysis and bertopic. *Humanities and Social Sciences Communications*, *11*, 603. https://doi.org/10.1057/s41599-024-03044-y

Klavans, R., & Boyack, K. W. (2009). Toward a consensus map of science. *Journal of the American Society for Information Science and Technology*, *60*(3), 455–476. https://doi.org/10.1002/asi.20991

Krauss, A. (2025). New scientific fields are triggered by powerful new methods. *Humanities and Social Sciences Communications*, *12*, 1590. https://doi.org/10.1057/s41599-025-05797-6

Leahey, E., Beckman, C. M., & Stanko, T. L. (2017). Prominent but less productive: The impact of interdisciplinarity on scientists' research. *Administrative Science Quarterly*, *62*(1), 105–139. https://doi.org/10.1177/0001839216665364

Li, D., & Agha, L. (2015). Big names or big ideas: Do peer-review panels select the best science proposals? *Science*, *348*(6233), 434–438. https://doi.org/10.1126/science.aaa0185

Liu, L., Wang, Y., Sinatra, R., Giles, C. L., Song, C., & Wang, D. (2018). Hot streaks in artistic, cultural, and scientific careers. *Nature*, *559*, 396–399. https://doi.org/10.1038/s41586-018-0315-8

Luo, X., Rechardt, A., Sun, G., et al. (2025). Large language models surpass human experts in predicting neuroscience results. *Nature Human Behaviour*, *9*(2), 305–315. https://doi.org/10.1038/s41562-024-02046-9

Marwitz, T., Colsmann, A., Breitung, B., et al. (2026). Predicting new research directions in materials science using large language models and concept graphs. *Nature Machine Intelligence*, *8*, 535–544. https://doi.org/10.1038/s42256-026-01206-y

Newman, M. E. J. (2001). The structure of scientific collaboration networks. *Proceedings of the National Academy of Sciences*, *98*(2), 404–409. https://doi.org/10.1073/pnas.98.2.404

Newman, M. E. J. (2006). Modularity and community structure in networks. *Proceedings of the National Academy of Sciences*, *103*(23), 8577–8582. https://doi.org/10.1073/pnas.0601602103

Park, M., Leahey, E., & Funk, R. J. (2023). Papers and patents are becoming less disruptive over time. *Nature*, *613*, 138–144. https://doi.org/10.1038/s41586-022-05543-x

Pedregosa, F., Varoquaux, G., Gramfort, A., et al. (2011). Scikit-learn: Machine learning in python. *Journal of Machine Learning Research*, *12*, 2825–2830.

Penner, O., Pan, R. K., Petersen, A. M., Kaski, K., & Fortunato, S. (2013). On the predictability of future impact in science. *Scientific Reports*, *3*, 3052. https://doi.org/10.1038/srep03052

Porter, A. L., & Rafols, I. (2009). Is science becoming more interdisciplinary? measuring and mapping six research fields over time. *Scientometrics*, *81*(3), 719–745. https://doi.org/10.1007/s11192-008-2197-2

Priem, J., Piwowar, H., & Orr, R. (2022). Openalex: A fully-open index of scholarly works, authors, venues, institutions, and concepts. *arXiv*. https://doi.org/10.48550/arXiv.2205.01833

Rafols, I., Porter, A. L., & Leydesdorff, L. (2010). Science overlay maps: A new tool for research policy and library management. *Journal of the American Society for Information Science and Technology*, *61*(9), 1871–1887. https://doi.org/10.1002/asi.21368

Reimers, N., & Gurevych, I. (2019). Sentence-bert: Sentence embeddings using siamese bert-networks. *Proceedings of EMNLP-IJCNLP 2019*, 3982–3992. https://doi.org/10.18653/v1/D19-1410

Saito, T., & Rehmsmeier, M. (2015). The precision-recall plot is more informative than the roc plot when evaluating binary classifiers on imbalanced datasets. *PLOS ONE*, *10*(3), e0118432. https://doi.org/10.1371/journal.pone.0118432

Schilling, M. A., & Green, E. (2011). Recombinant search and breakthrough idea generation: An analysis of high impact papers in the social sciences. *Research Policy*, *40*(10), 1321–1331. https://doi.org/10.1016/j.respol.2011.06.009

Shannon, C. E. (1948). A mathematical theory of communication. *Bell System Technical Journal*, *27*(3), 379–423. https://doi.org/10.1002/j.1538-7305.1948.tb01338.x

Shi, F., & Evans, J. A. (2023). Surprising combinations of research contents and contexts are related to impact and emerge with scientific outsiders from distant disciplines. *Nature Communications*, *14*, 1641. https://doi.org/10.1038/s41467-023-36741-4

Sinatra, R., Wang, D., Deville, P., Song, C., & Barabasi, A.-L. (2016). Quantifying the evolution of individual scientific impact. *Science*, *354*(6312), aaf5239. https://doi.org/10.1126/science.aaf5239

Stirling, A. (2007). A general framework for analysing diversity in science, technology and society. *Journal of the Royal Society Interface*, *4*(15), 707–719. https://doi.org/10.1098/rsif.2007.0213

Torres-Salinas, D., & Arroyo-Machado, W. (2026). The 'big three' of scientific information: A comparative bibliometric review of web of science, scopus, and openalex. *arXiv*. https://doi.org/10.48550/arXiv.2601.21908

Traag, V. A., Waltman, L., & van Eck, N. J. (2019). From louvain to leiden: Guaranteeing well-connected communities. *Scientific Reports*, *9*, 5233. https://doi.org/10.1038/s41598-019-41695-z

Uzzi, B., Mukherjee, S., Stringer, M., & Jones, B. (2013). Atypical combinations and scientific impact. *Science*, *342*(6157), 468–472. https://doi.org/10.1126/science.1240474

van Eck, N. J., & Waltman, L. (2010). Software survey: Vosviewer, a computer program for bibliometric mapping. *Scientometrics*, *84*(2), 523–538. https://doi.org/10.1007/s11192-009-0146-3

Virtanen, P., Gommers, R., Oliphant, T. E., et al. (2020). Scipy 1.0: Fundamental algorithms for scientific computing in python. *Nature Methods*, *17*, 261–272. https://doi.org/10.1038/s41592-019-0686-2

Visser, M., van Eck, N. J., & Waltman, L. (2021). Large-scale comparison of bibliographic data sources: Scopus, web of science, dimensions, crossref, and microsoft academic. *Quantitative Science Studies*, *2*(1), 20–41. https://doi.org/10.1162/qss_a_00112

Waltman, L., & van Eck, N. J. (2012). A new methodology for constructing a publication-level classification system of science. *Journal of the American Society for Information Science and Technology*, *63*(12), 2378–2392. https://doi.org/10.1002/asi.22748

Wang, D., Song, C., & Barabasi, A.-L. (2013). Quantifying long-term scientific impact. *Science*, *342*(6154), 127–132. https://doi.org/10.1126/science.1237825

Wang, H., Fu, T., Du, Y., et al. (2023). Scientific discovery in the age of artificial intelligence. *Nature*, *620*, 47–60. https://doi.org/10.1038/s41586-023-06221-2

Wang, J., Veugelers, R., & Stephan, P. (2017). Bias against novelty in science: A cautionary tale for users of bibliometric indicators. *Research Policy*, *46*(8), 1416–1436. https://doi.org/10.1016/j.respol.2017.06.006

Wu, L., Wang, D., & Evans, J. A. (2019). Large teams develop and small teams disrupt science and technology. *Science*, *363*(6422), 378–382. https://doi.org/10.1126/science.aao9021

Wuchty, S., Jones, B. F., & Uzzi, B. (2007). The increasing dominance of teams in production of knowledge. *Science*, *316*(5827), 1036–1039. https://doi.org/10.1126/science.1136099

Yegros-Yegros, A., Rafols, I., & D'Este, P. (2015). Does interdisciplinary research lead to higher citation impact? the different effect of proximal and distal interdisciplinarity. *PLOS ONE*, *10*(8), e0135095. https://doi.org/10.1371/journal.pone.0135095